Gayathri Gopalan

BI & Data Analytics Portfolio

6+ years of experience transforming raw data into business decisions. This portfolio showcases four end-to-end analytical projects spanning Power BI dashboards, Tableau, NLP/ML in Python, and time-series forecasting — each structured from business problem through methodology to measurable outcome.

01 · Agent Recruitment Dashboard

Power BI · SQL · Excel

Built a drill-down Power BI dashboard for a financial services firm to analyse agent conversion rates across 7,179 applicants, surfacing the demographic and qualification profiles that drove business outcomes.

02 · Global Suicide Rate Dashboard

Tableau · Excel · WHO Data

Designed an interactive Tableau dashboard over 37 years of WHO mortality data, enabling public health policymakers to explore suicide rates by gender, age group, and country across three decades.

03 · COVID-19 Vaccine Sentiment

Python · NLP · Tweepy · Random Forest

Scraped live Twitter data via Tweepy API and built an NLP sentiment classifier to measure public perception of vaccine brands during the 2021 rollout — delivering actionable communication insights.

04 · Tesla Stock Price Prediction

Python · ARIMA · LSTM Neural Network

Compared classical ARIMA(3,2,2) vs. LSTM deep learning on 10 years of TSLA price data, demonstrating that LSTM reduced forecast error by >50% on high-volatility, non-linear equity movements.

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PROJECT 01

Agent Recruitment Analysis Dashboard

Power BI · SQL · Excel

Business Problem

A financial services company (FINMAN) needed to understand their agent recruitment pipeline. Of 7,179 applicants, only 2,388 (33%) successfully sourced business. Leadership lacked visibility into *which demographic segments* converted best and *why* two-thirds of recruits failed to generate revenue — preventing data-driven HR decisions.

Analytical Approach

- Queried raw recruitment data via SQL; built a clean joined dataset linking demographics, occupation, qualification, and sourcing outcomes.
- Designed a Power BI star-schema data model with calculated measures for conversion rates by manager grade, age bin, and occupation type.
- Delivered an interactive drill-down dashboard with cross-filter capability — click any segment to isolate its full performance profile.

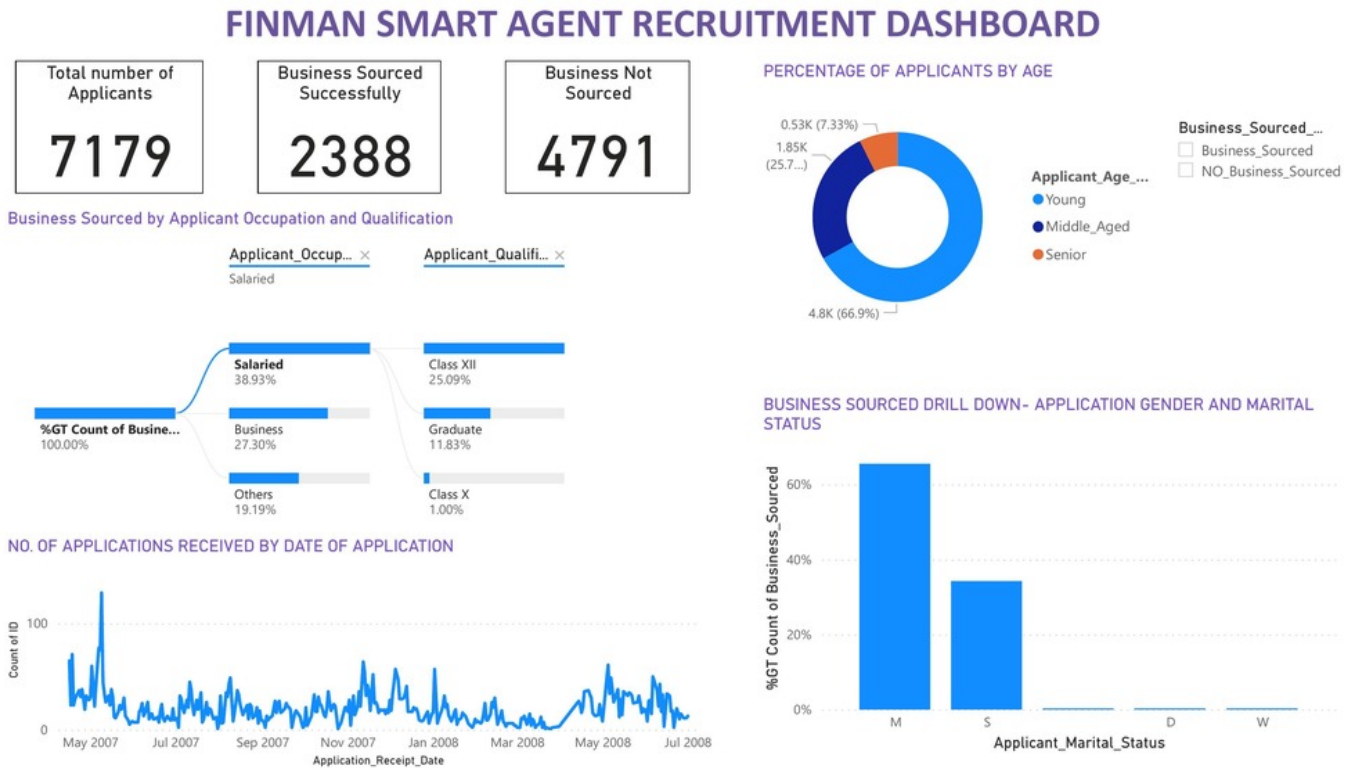


Fig 1 — Overview: 7,179 applicants, 33% conversion. Sankey maps occupation → qualification flow. Donut: 66.9% of applicants are young.



Fig 2 — Drill-down: Manager grade 7–9 hit 50–57% conversion vs grade 1 at 12.5%. Recruitment peaked 2006 then sharply declined.

Business Impact

■ Salaried + graduate profile = highest conversion → HR can now target this segment to maximise recruitment ROI.

■ Grade 1 managers at 12.5% identified as highest-priority coaching cohort; Grade 7–9 set as internal benchmark.

■ 2006–2008 pipeline decline surfaced proactively — enabling leadership to act before it became a revenue problem.

■ Middle-aged manager outperformance vs. young applicant majority signals a structural onboarding gap.

PROJECT 02

Global Suicide Rate Analysis Dashboard

Tableau · Excel · WHO Dataset 1979–2016

Policy Problem

WHO tracks 37 years of suicide data across 100+ countries and 6 age groups — but raw tabular data is not actionable for policymakers. The goal was to surface *who* is most at risk, *where* rates are highest, and *how* trends shifted across decades, enabling health agencies to prioritise prevention resource allocation.

Analytical Approach

- Cleaned 37 years of WHO mortality records in Excel; loaded into Tableau with per-100k rate fields.
- Built a multi-view dashboard: gender ratio, age-group breakdowns, country scatter plots, and a 3-decade trend line.
- Parameter controls enable dynamic filtering by year, country, gender, and age group.

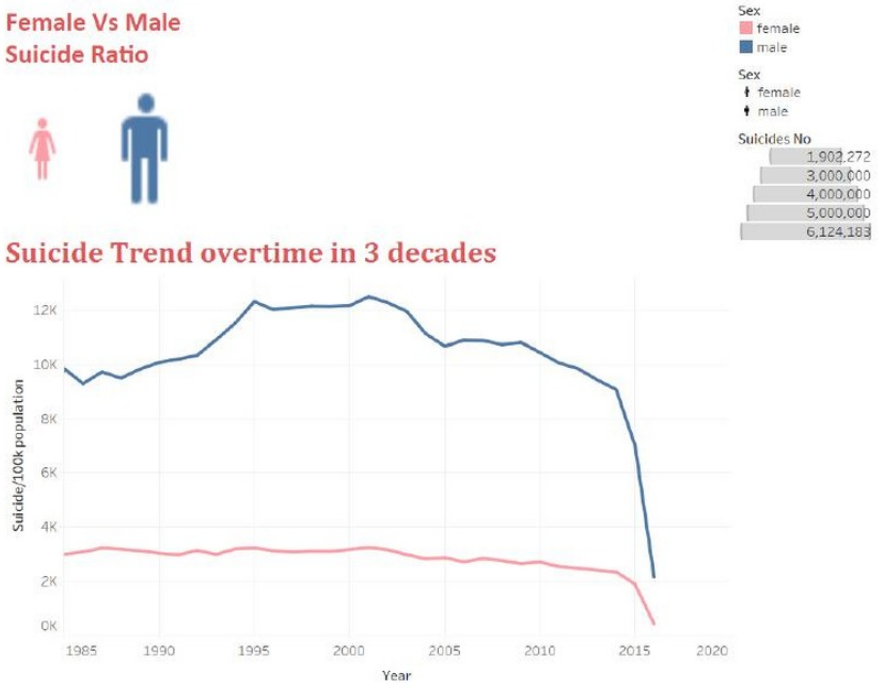


Fig 3 — Female vs Male ratio and 3-decade trend: male rates consistently ~4× female. Global rates peaked ~2000 then declined sharply.

➔ **Age-wise Suicide statistics**

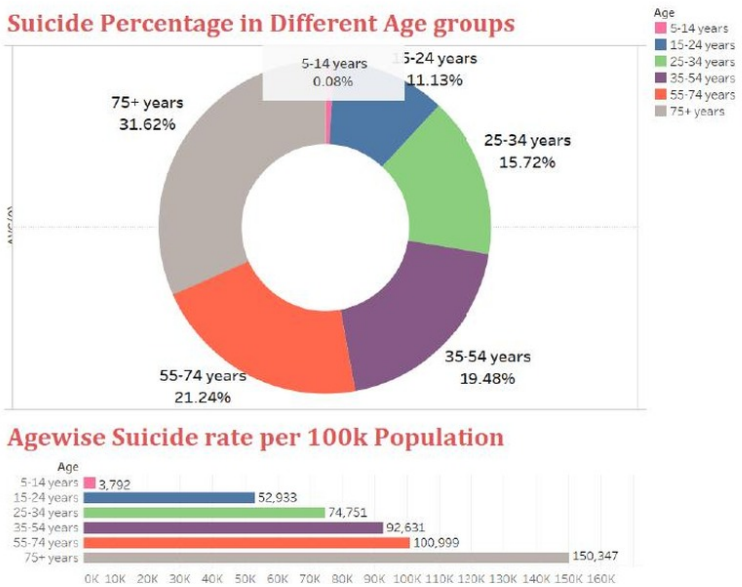


Fig 4 — Age breakdown: 75+ holds 31.6% of suicides and highest rate per 100k (150,347) — 3× the 35–54 group.

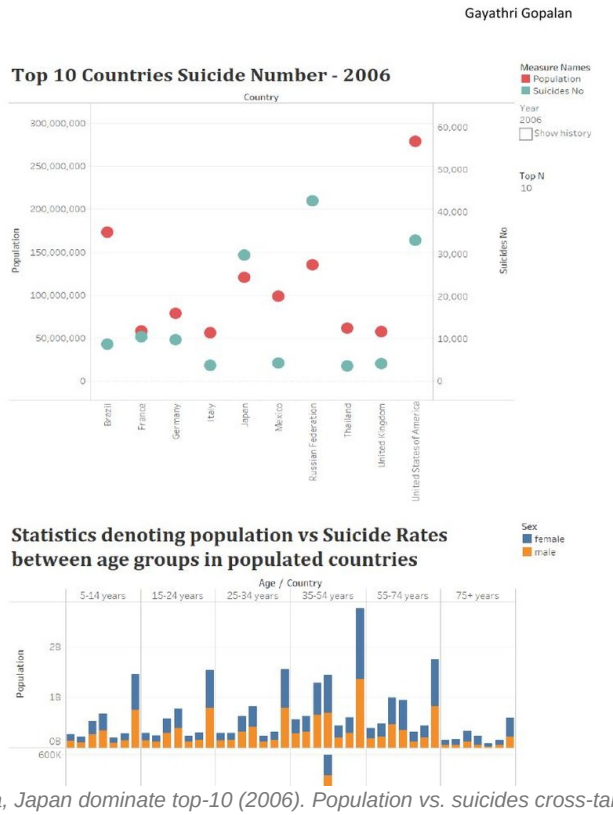


Fig 5 — Tableau dashboard: USA, Russia, Japan dominate top-10 (2006). Population vs. suicides cross-tab highlights Russia/UK disparity.

Key Policy Insights

■ Males = ~76% of all suicides globally — the single highest-leverage target for any prevention programme.

■ Rates declined post-2001 despite population growth — measurable evidence that public health campaigns deliver impact.

■ 75+ age group critically under-served: highest per-capita risk, rarely the focus of mental health outreach.

■ Dashboard turns 37 years of static WHO data into a live analyst tool filterable across any dimension.

PROJECT 03

COVID-19 Vaccine Sentiment Analysis

Python · Tweepy (Twitter API) · NLP / NLTK · Random Forest Classifier

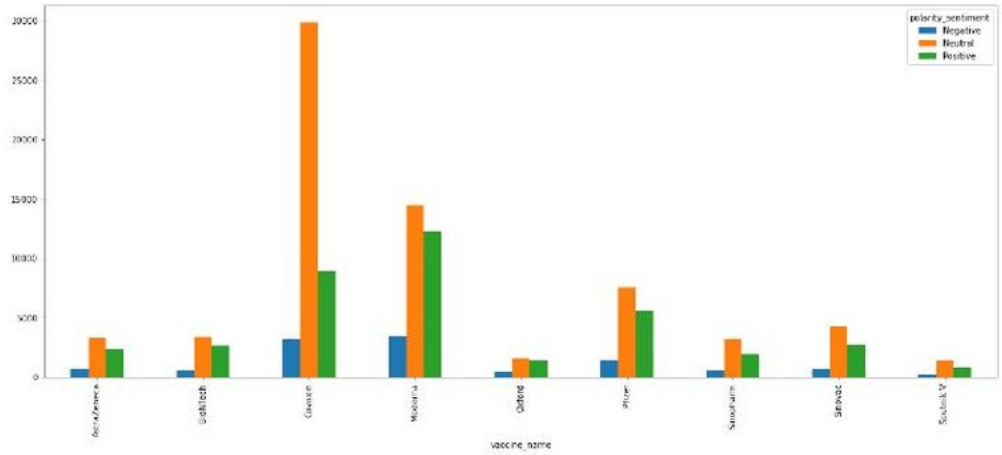
Business Problem

During the 2021 rollout, public health agencies needed real-time intelligence on sentiment toward specific vaccines (Covaxin, Moderna, Pfizer, Sputnik V, Sinovac). Identifying which vaccines were driving negative perception could directly shape communication strategy and counter vaccine hesitancy.

Technical Approach

- Collected live tweets via Tweepy API using vaccine-name keyword queries; saved as structured CSV with user metadata and engagement metrics.
- NLP pipeline: lemmatization → tokenization → lowercase normalisation → stopword removal → feature extraction.
- Trained Random Forest Classifier on Negative / Neutral / Positive sentiment classes; validated on hold-out test set.
- Visualised with vaccine-level sentiment bar, overall polarity distribution, and top-500-term word cloud.

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plot_size[1] = 10
plot.rcParams["figure.figsize"] = plot_size
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From the bar chart visualization, we can find that Covaxin, Moderna and Pfizer are the vaccines

Fig 6 — Sentiment by vaccine brand: Covaxin and Moderna dominate tweet volume. All vaccines show Neutral/Positive majority.

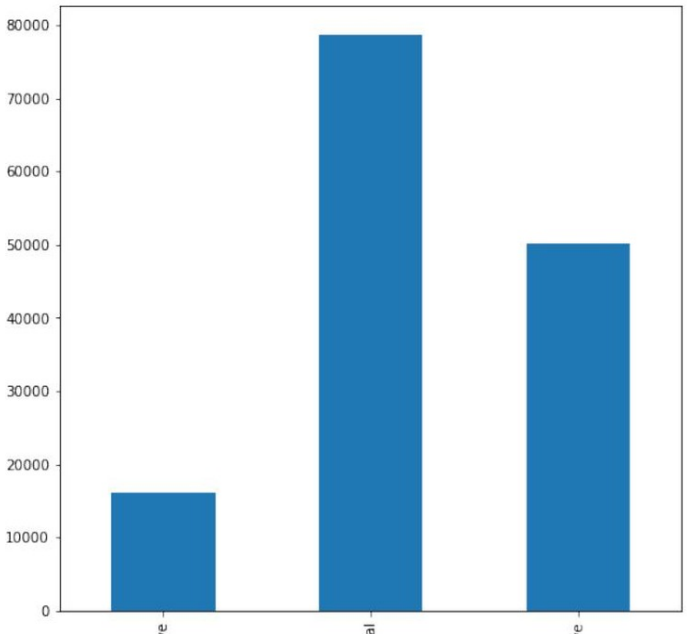


Fig 7 — Overall polarity: 78K Neutral, 50K Positive, only 16K Negative (~11%). Rollout perception was broadly favourable.

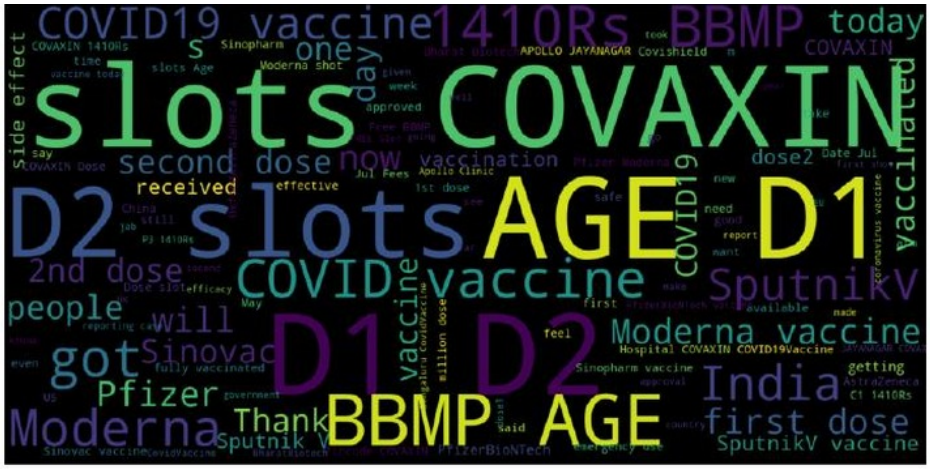


Fig 8 — Word cloud: "slots", "dose", "age", "D1/D2" dominate — public concern was access logistics, not safety.

Business Value

■ Negative sentiment at 11% gave agencies evidence-based reassurance that communication strategy was working.

■ Slot availability and age eligibility were top concerns — pointing exactly where FAQ content should be focused.

■ Sputnik V/Sinovac showed low volumes, flagging regional monitoring gaps.

■ Pipeline operationalisable as real-time sentiment dashboard for any public health or product launch campaign.

PROJECT 04

Tesla Stock Price Prediction (ARIMA vs LSTM)

Python · ARIMA (statsmodels) · LSTM Neural Network (Keras) · Matplotlib

Analytical Problem

Financial analysts need forward-looking price signals for trading and risk management. Using 10 years of TSLA daily market data (2010–2021), the objective was to compare classical statistical ARIMA vs. deep learning LSTM RNN and determine which delivers more reliable predictions for a high-volatility, non-linear asset.

Modelling Approach

- EDA on 10 years of TSLA daily open/close prices; applied 8-day rolling moving average to smooth noise without losing trend.
- ADF stationarity test → 2nd-order differencing → ARIMA(3,2,2) fitted via maximum likelihood on 2,687 observations.
- Time series decomposed into trend, seasonal, and residual components to understand structural drivers.
- LSTM RNN trained over 200 epochs (95/5 split); both models evaluated by RMSE on the volatile 2020–2021 test window.

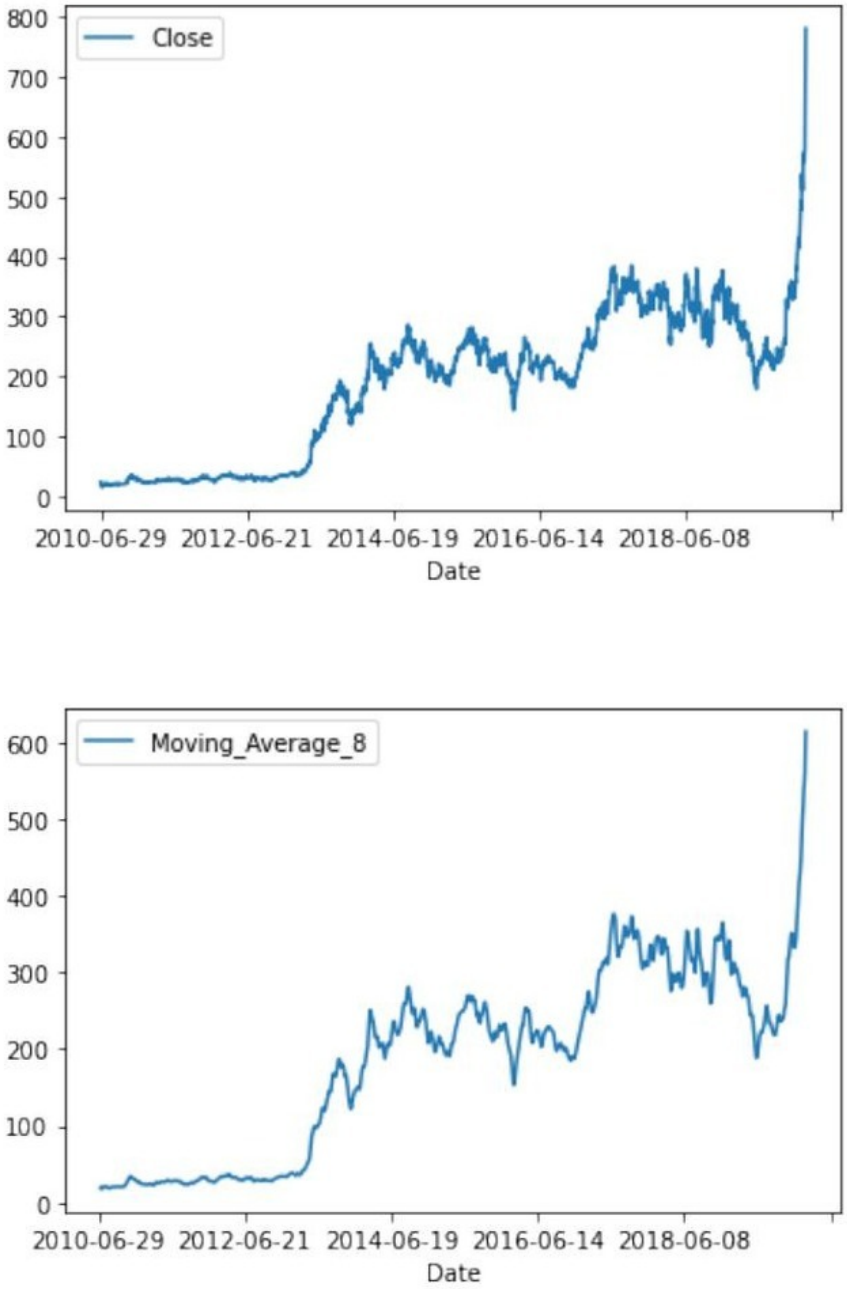


Fig 9 — TSLA close price (2010–2020) with 8-day moving average (both charts). Exponential non-linear trajectory signals classical models will underperform.



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Fig 10 — LSTM predictions: training fit (red) tracks original data closely; test predictions (yellow) capture the 2020–2021 upward surge.

Results & Recommendations

- ARIMA error rate 19.39% — useful for directional signals but too imprecise for tactical decisions.
- LSTM delivered significantly lower RMSE (~6 train, ~11 test), accurately tracking the non-linear 2020–2021 explosion.
- Conclusion: For high-volatility equities, LSTM outperforms ARIMA — deep learning is the recommended forecasting approach.
- Pipeline reusable across any equity time series; extensible with sentiment feeds, macro indicators, or transformer architectures.